

A REVIEW AND MASTERY BLOCK FOR

Theorems 1–16

Augros, Introduction to Arithmetic and Geometry, Chapter 1

A Direct Instruction consolidation block, a permanent cumulative warm-up,
and notes on running a classroom of two

*Prepared for a homeschool classroom of two students, ages nine and twelve, working through the
demonstrations by hand.*

How to use this packet. Parts One through Five are the design — read them once. Part Six is the twelve-day block; run it with the packet open on the table. Part Seven is what you keep doing forever afterward. Part Eight is how you know it worked. Part Nine is about you rather than about geometry, and is the part most likely to determine whether any of the rest survives contact with a real year.

Part One · Why the Thirty Percent Happens

1.1 The diagnosis

Thirty percent recall is not a discipline problem and not an ability problem. It is the ordinary arithmetic of unrehearsed exposure. A proof met once and never met again decays on a known schedule, and after a few weeks a learner is holding roughly what your two are holding. Nothing you have done is wrong. You have built the first half of the machine — the half that presents material well. The second half, the half that brings material back, does not exist yet.

The other symptom you describe is the more diagnostic one, and it points somewhere specific. A student loses his place inside a proof when the proof's *components* are not automatic. Every step of a Euclidean argument asks him to do three things at once: recognise a configuration, recall the authority that licenses the step, and hold the goal of the proof in mind. If recalling the authority costs effort — if “why are those two lines equal?” requires a search rather than a reflex — then the search consumes the very capacity that was holding the goal, and the goal drops. He is not lost because he forgot the proof. He is lost because he spent his place-holding capacity on retrieval.

THE CONSEQUENCE FOR REVIEW

Do not review by re-reading proofs. Re-reading whole proofs at low component fluency reproduces exactly the condition that caused the problem in the first place — it feels like learning, because recognition feels like knowledge, and it moves nothing.

Firm the components to the point of reflex, teach the small number of strategies explicitly *as strategies*, and the proofs largely reassemble themselves. A Euclidean proof is not a memory object. It is a performance built out of parts.

1.2 The three layers underneath every proof

Everything in Theorems 1–16 sits on three layers. Almost all review works only on the top layer, the proofs themselves.

Layer	What it contains	How many	Target performance
1 · The Givens	Definitions, postulates, common notions — the authorities a step may cite	36 total — 14 do the real work	Cited on sight, under a second, without hesitation
2 · The Operations	The nine recurring actions: draw the circle, join, extend, cut off, bisect, prove a congruence	9	Given a figure and a goal, names the operation it calls for
3 · The Ways of Demonstrating	Construct and verify · discover congruences · reductio · find what is equal · trichotomy elimination	5	Names the way from the bare statement, before drawing anything

Layer	What it contains	How many	Target performance
+ The order of dependence	Which theorem rests on which	—	Can point to where we are, what we stand on, and what will stand on us

Knowing the order of dependence is the direct answer to losing the place. A student who cannot say what a proof is *for* — what it will be used to prove later — has no structure to hold the steps in, and so the steps are sixteen unrelated stories. The wall chart shows them as one connected order.

1.3 What “mastery” will mean here

“Mastery” is unusable as a target until it is behavioural. Four tiers, each one testable in under a minute:

Tier	The student can...	How you test it
1 · Recall	Give the theorem from its number, and the number from its statement	Two-way deck, oral, one second
2 · Classify	Name the way of demonstrating, and how the theorem is proved	“Theorem 4 — which way? proved by what?”
3 · Reconstruct	Give the whole proof orally from the bare diagram, citing every reason	Diagram-only reconstruction
4 · Cold write	Write the demonstration from a blank page, no diagram supplied, three or more days after last exposure	Timed, silent, no prompts

Your two are somewhere between Tier 1 and Tier 2 on most of 1–16 right now — which is precisely what “they enjoy it and remember about thirty percent” describes.

SET THE TARGET HONESTLY

Tier 3 on all sixteen. Tier 4 on the keystone six (Theorems 1, 2, 6, 8, 11, 14).

Tier 4 on all sixteen is not worth what it costs, and chasing it is the fastest way to make the geometry hateful. A boy who can reconstruct any of the sixteen from its diagram, and can write six of them cold, owns Chapter 1 in every sense that matters for Chapter 2.

Part Two • The Order of Dependence

Print the wall chart that accompanies this packet and put it where the boys can reach it. Every session begins and ends with somebody's finger on it. The chart is not decoration; it is what turns sixteen separate proofs into one connected order, and it is the cheapest fix available for losing the place.

2.1 Four stages

Chapter 1's first sixteen theorems are not a list. They fall into four stages, each with a job, and each one built entirely out of the stage before it.

Stage	Theorems	The job it does	What it turns on
I • Equality without measuring	1–2	Get two things known to be equal without ever laying a ruler on them, and get the first congruence test	A circle hands you equality for free (Post. 3 + Def. 17). Superposition hands you SAS.
II • The congruence toolkit	3–6	Build the two instruments — SAS and SSS — that will prove nearly everything afterwards	Isosceles triangles first, then Theorem 5. SSS is not proved directly: it follows from Theorem 5.
III • The construction toolkit	7–10	Learn to bisect and to drop perpendiculars — the operations later proofs will need as moves	Every one of them is the equilateral triangle of Thm. 1, verified by SSS or SAS.
IV • Straight lines, then inequality	11–16	Angles on a straight line; then, for the first time, prove something is <i>greater</i> rather than equal	Two right angles as a fixed amount to measure against; then Theorem 14, which every later inequality rests on.

Teach these four names to the boys and use them out loud. “We are in the construction toolkit” locates a proof in a way that “Theorem 9” never will.

2.2 The keystone six

Six of the sixteen carry the weight. If these are at Tier 4, most of the rest can be reconstructed on demand even when the details have gone; if any of these is soft, everything downstream of it wobbles.

Keystone	What it is	What it feeds
Theorem 1	Equilateral triangle on a given line	7, 8, 9 — and through 8, also 10 and 14
Theorem 2	SAS	3, 4, 8, 14 — the most-cited theorem in the chapter
Theorem 6	SSS	7, 9, 10 — every construction's verification
Theorem 8	Bisect a straight line	10, 14
Theorem 11	Adjacent angles on a line = two right angles	12, 13 — and 13 feeds 14

Keystone	What it is	What it feeds
Theorem 14	Exterior angle > remote interior	15, then 16 — the whole of Stage IV.

THEOREM 14 DESERVES ITS OWN DAY

Theorem 14 uses more earlier results than any proof before it. It bisects a side (Thm. 8), joins and doubles a line out to a new point (Post. 1, Post. 2), uses vertical angles (Thm. 13), applies SAS (Thm. 2), and finishes on whole-greater-than-part (C.N. 5) — and then does the whole thing a second time for the other remote angle.

If your boys are losing the place anywhere in 1–16, the odds are very good that it is here, and that the softness shows up later as an inability to reconstruct 15 and 16. Day 10 of the block is Theorem 14 alone.

2.3 The converse pairs — and the most valuable insight in the chapter

Three of the sixteen are converses of others, and the pairing is the best organising idea available to a student trying to hold the chapter in his head.

Forward	Converse	How the forward one is proved	How the converse is proved
3 — isosceles $\Rightarrow$ equal base angles	4 — equal base angles $\Rightarrow$ isosceles	Directly: construct, then discover two congruences	By contradiction (C.N. 5)
11 — line on a line $\Rightarrow$ two right angles	12 — two right angles $\Rightarrow$ straight line	Directly: construct a perpendicular, add the pieces	By contradiction (C.N. 5)
15 — greater side $\Rightarrow$ greater angle	16 — greater angle $\Rightarrow$ greater side	Directly: cut off, isosceles, exterior angle	By elimination of the other two cases

The pattern is not an accident, and it is worth stating to them as a law of the terrain: the converse always costs more than the theorem, and it is nearly always paid for in indirect proof. A student who knows this does not merely remember Theorem 16 — he could have predicted its shape from its statement.

Two further pairs are worth naming even though they are not converses: Theorems 9 and 10 are the same construction for a point *on* the line and a point *off* it, and Theorems 2 and 6 are the two congruence tests, one bought by superposition and one bought by rigidity.

2.4 The order of dependence, in full

This is the content of the wall chart, in table form, for your own reference while building drills. “Rests on” lists only the theorems, not the definitions and postulates.

#	Theorem	Rests on	Is used by
1	Equilateral triangle	—	7, 8, 9
2	SAS	—	3, 4, 8, 14
3	Isosceles base angles	2	5, 15, 16
4	Converse of isosceles	2	—
5	Triangles are rigid	3	6
6	SSS	5	7, 9, 10
7	Bisect an angle	1, 6	8
8	Bisect a line	1, 7, 2	10, 14
9	Perpendicular from a point on a line	1, 6	11
10	Perpendicular from a point off a line	8, 6	—
11	Adjacent angles = two rights	9	12, 13
12	Converse of 11	11	—
13	Vertical angles equal	11	14
14	Exterior angle > remote interior	8, 13, 2	15
15	Greater side $\Rightarrow$ greater angle	3, 14	16
16	Greater angle $\Rightarrow$ greater side	3, 15	—

Notice what the last column tells you: nothing later in these sixteen depends on Theorems 4, 10, 12 or 16. That is not a reason to neglect them — 10 and 16 both return in later chapters — but it is useful when a session is short. When you have seven minutes, spend them on a keystone, not on a theorem nothing else rests on.

Part Three • The Components to Firm

3.1 The fourteen givens that do the real work

Augros gives twenty-four definitions, five postulates, five common notions and two extra principles. Thirty-six items is too many to hold at reflex speed — and it is not necessary. **Every single citation of a definition, postulate or common notion in Theorems 1–16 is one of the fourteen below.** Drill these fourteen to automaticity first; the remaining definitions are vocabulary, and can be drilled lightly, for recognition rather than speed.

The item	What it licenses you to say	Where it appears in 1–16
Post. 1 — join	“Join A and C.”	1, 3, 4, 5, 7, 9, 10, 14, 15 — the most-cited authority of all
Post. 2 — extend	“Produce BC to D.”	12, 14 — and the extension in 3
Post. 3 — circle	“Draw a circle about A with radius AB.”	1, 3, 4, 7, 9, 10
Post. 4 — all right angles are equal	Compare two “two right angles” quantities	12 — its one essential use
Def. 9 — right angle	Call two equal adjacent angles <i>right</i>	9, 10, 11
Def. 10 — perpendicular	Call the line perpendicular	9, 10
Def. 17 — radius	“These are equal, being radii of the same circle.”	1 and 10 by name — and in force wherever a circle cuts off an equal length (3, 4, 7, 9, 15)
Def. 20 — equilateral / isosceles	Name the triangle and read off equal sides	1, 3, 7, 8, 9
C.N. 1 — transitivity	“Each equals the same thing, so they equal each other.”	1, 11, 12, 13
C.N. 2 — add equals	“Add equals to equals.”	3
C.N. 3 — subtract equals	“Take equals from equals.”	3, 13
C.N. 4 — coincidence	“Things that coincide are equal.”	2, 6
C.N. 5 — whole > part	Every inequality, and most contradictions	4, 5, 12, 14, 15
Two lines cannot enclose a space	Force the third side to coincide	2

TWO THINGS TO SAY OUT LOUD, REPEATEDLY

Postulate 3 with Definition 17 does more work than anything else in the chapter. A circle is how you get equality without measuring anything: Postulate 3 draws it, Definition 17 tells you the radii are equal. Six of the sixteen proofs open by drawing a circle, and several more cut off an equal length, which is the same move under another name. (The single most-cited authority, incidentally, is Postulate 1 — joining two points — precisely because it is so easy to do without noticing that it needed a licence.)

Common Notion 5 is used for two different jobs. It is how you prove that something is *greater* (Theorems 14, 15), and it is how you reach nearly every contradiction (Theorems 4, 5, 12). Teach it as the most important sentence in the chapter: when a boy is stuck on a “greater than,” or stuck inside a *reductio* and can’t find the absurdity, the answer is almost always Common Notion 5. That single heuristic will unstick him more often than anything else you can give him.

3.2 The nine operations

An operation is an action on the diagram together with the authority that permits it. Students who cannot construct a proof usually can, in fact, carry out every operation in it — they simply do not have the nine written down as a list to choose from, so they stare at the figure instead of working down the list. Make the list explicit.

	The operation	Permitted by	Used in
O1	Draw a circle to get equal lines	Post. 3 + Def. 17	1, 3, 4, 7, 9, 10
O2	Join two points	Post. 1	1, 3, 4, 5, 7, 9, 10, 14, 15
O3	Extend a line past its end	Post. 2	12, 14 — and 3
O4	Cut off a length equal to a given one	Post. 3 (circle)	3, 4, 15
O5	Erect the equilateral triangle on a segment	Thm. 1	7, 8, 9
O6	Bisect — an angle, or a line	Thm. 7 / Thm. 8	8, 10, 14
O7	Discover a congruence: prove two triangles equal, then take the corresponding part you wanted	Thm. 2 (SAS) or Thm. 6 (SSS)	3, 6, 7, 8, 9, 10, 14
O8	Add equals, subtract equals, or chain equals	C.N. 1, 2, 3	1, 3, 11, 12, 13
O9	Whole greater than part	C.N. 5	4, 5, 12, 14, 15

Operation 7 is the most used of the nine. Nearly every proof of an equality in Chapter 1 is: find two triangles, prove them equal, take the corresponding part. The step students omit is naming the triangles *in corresponding order* — $\triangle ADC$ and $\triangle AEB$, not “those two triangles.” Insist on the order; that is where the reasoning is.

3.3 The five ways of demonstrating

This is Layer 3, and it is the layer that most geometry teaching never states explicitly at all. There are exactly five ways of demonstrating in Theorems 1–16. A student who can name the way from the bare

statement — before he has drawn anything — has most of the proof already, because the way determines the shape of the argument.

Way A • Construct and Verify

Theorems 1, 7, 8, 9, 10. Shape: build the figure with circles and lines; then *prove* that what you built does what you claimed, always by SAS or SSS.

Tell: the statement begins “Given ... construct ...”.

The step they will skip: the verification. They build it and stop, because building it *looks* like it worked. Half the demonstration is showing that it did.

Way B • Discover Congruences

Theorems 3, 6, 14 — and the second half of every Way A proof. Shape: prove two triangles equal by SAS or SSS, then take the corresponding part you were after all along.

Tell: the goal is an equality of two segments or two angles.

The step they will skip: naming which triangles, in corresponding order, and naming which test. “They’re congruent” is not a step.

Way C • Reductio ad Absurdum

Theorems 4, 5, 12. Shape: assume the contrary; run it forward honestly; land on a whole equal to its own part, or an angle equal to itself plus something more; declare the absurdity and conclude.

Tell: the statement is a converse, or asserts that something is impossible.

The step they will skip: ruling out the *other* direction. In Theorem 4 they show AB is not greater than AC and stop — but AC not greater than AB is a second case, and without it the trichotomy is not closed.

Way D • Find What Is Equal

Theorems 11, 12, 13. Shape: find two quantities that are each equal to “two right angles,” or each equal to some common thing; conclude they are equal by C.N. 1, then take away a common part by C.N. 3.

Tell: the phrase “two right angles” appears in the statement.

The step they will skip: in Theorem 11, noticing that the perpendicular is *constructed* by Theorem 9, not handed to them. They treat it as given, and then cannot say why it exists.

Way E • Trichotomy Elimination

Theorem 16. Shape: exactly three cases are possible — greater, equal, less. Kill two of them with theorems already proved. The survivor is the conclusion.

Tell: the converse of an inequality theorem.

The step they will skip: they attempt a direct construction, because every proof before it had one. Theorem 16 has no construction at all — it is pure logic on top of Theorems 3 and 15, and saying so out loud is most of the teaching.

CONNECTIONS TO THE LOGIC COURSE — SPEND A SESSION ON THIS

You are teaching Aristotelian logic to the same two boys. Chapter 1 is an unusually good source of live specimens, and using it that way makes each subject cheaper to remember.

Converses and illicit conversion. “Every isosceles triangle has equal base angles” is an A-proposition, and A-propositions do not convert simply. Theorem 4 is what that costs — an entire second demonstration, by contradiction, to buy back what a careless student would have assumed for free. You will not get a better example of illicit conversion in your logic course than one the boys have personally paid for in geometry. The same specimen appears twice more, at 11/12 and at 15/16.

Reductio. Way C is *reductio ad impossibile*, which is the instrument Aristotle himself uses to validate Baroco and Bocardo. The boys can meet the same tool twice in one week, in two subjects, and recognise it the second time.

Every step is an enthymeme. “AC and AB are radii of the same circle, therefore $AC = AB$ ” is a syllogism with its major premise suppressed — and the suppressed major is exactly the citation. Requiring them to supply the authority on every single step is logic practice, and it is simultaneously the precise fluency the geometry needs. This is the deepest reason the citation drill in Part Five is worth its time.

Part Four · The Direct Instruction Mechanics

You already know these from the logic course. This part is only about how they land on geometry, where the unit of instruction is a demonstration rather than a rule.

4.1 Model — Lead — Test

The single most important change to how you present a proof. Most teaching is Test with a running commentary: the teacher asks questions all the way through, the students supply fragments, and nobody ever hears or performs the argument whole. Separate the three phases and do not blend them.

Phase	Who does it	The rule
MODEL	You alone	Do the entire proof at the board, aloud, citing every reason. Ask no questions. They watch and listen. If you ask questions here you have not modelled anything.
LEAD	All three of you, in unison	You draw; all of you say the proof together, out loud, at the same time. Both boys speak every step. This is the phase people skip, and it is where the language gets into their mouths.
TEST	One student alone	He does it; you say nothing at all except “Reason?” when he asserts a step without an authority. No hints, no faces, no finishing his sentences.
DELAYED TEST	One student alone, a day or more later	The same proof, cold, at the start of a later session. This is the only phase that measures anything real.

Model. “My turn. Watch my hands and listen for the reasons.”

[Do all of Theorem 8, aloud, start to finish, no questions.] “Make the equilateral triangle ABC on AB — by Theorem 1. Bisect the angle ACB with CD — by Theorem 7. AC equals CB — Definition 20, equilateral. Angle ACD equals angle BCD — because CD bisects it. CD is common to both triangles. So by Side-Angle-Side, Theorem 2, the triangles are equal. Therefore AD equals DB, and D is the midpoint. Q.E.D.”

Lead. “Together. My hands, all our voices.” [Redraw; all three say every step in unison.] Do this twice.

Test. “Your turn, alone. Board, please.” [Silence. If he says “AC equals CB” with no citation: “Reason?” — nothing else.]

Delayed test. Tomorrow, first thing: “Theorem 8, from the diagram, cold.”

4.2 Signals and choral response

With a class of two, unison response is worth more than it is in a class of twenty-five, for a reason that is specific to your situation: it guarantees that the nine-year-old answers every single question out loud, and it prevents the twelve-year-old from being the one who always gets there first. It also converts a forty-five-minute session into two or three hundred individual responses rather than thirty, which is most of where the retention comes from.

- **Pick one signal and never vary it.** A hand held up while you ask, dropped when you want the answer. The pause between question and drop is the thinking time — vary its *length*, never its form.
- **Ask, pause, signal, respond.** If you signal before the pause, the fast one answers and the slow one echoes. The pause is the whole mechanism.
- **Listen for two voices.** A unison answer that is one voice plus an echo half a beat behind is a failed item, even if the answer was right. Repeat it.
- **Choral first, individual second, always.** Every item gets a unison turn before anyone gets a solo turn. The solo turn is a check, not the teaching.
- **Individual turns: youngest first on new material, oldest first on old.** On new material this stops the younger boy from being permanently second; on old material it lets him hear a fluent model of something he has already met.

4.3 The firming cycle

What you do at the exact moment of an error. It is five steps, it takes about twenty seconds, and steps 3 and 5 are the ones everybody drops — which is why errors keep coming back.

The boy cannot say why two radii are equal.

1 • Correct — immediately, with no explanation. “Definition seventeen.” Do not explain here. Explanation at the moment of error teaches him that a wrong answer buys a lecture.

2 • Repeat the same item. “My turn: Definition seventeen. Your turn — why are AC and AB equal?” → “Definition seventeen.”

3 • Back up three items and run forward through the missed one again. This is the step that makes it stick. The item has to be answered correctly *in the middle of a stream*, not in isolation, or what he has learned is only how to answer it when it is flagged.

4 • End of the session, cold. Ask it once more before you dismiss.

5 • First item in tomorrow's warm-up. Write it in the log now, or step 5 will not happen.

Three things never to do at the moment of error: explain; let the other boy answer for him; or move on with the error standing. The third is the expensive one — an uncorrected error in Theorem 6 is an unteachable Theorem 7.

WHY THIS MATTERS MORE FOR A PARENT THAN FOR A TEACHER

When the correction is a fixed five-step procedure, it carries no displeasure. You are not disappointed in him; you are executing step two. That is the underrated benefit of Direct Instruction in a home classroom — the script absorbs the emotional weight of correction, so that being wrong in front of one's father costs nothing. Wrong answers offered freely are the entire diagnostic instrument you have. Protect the supply.

4.4 The mastery criterion, and delayed testing

The 90/100 rule. Move forward when first-try accuracy on the day's drill is about ninety percent and repeat accuracy is one hundred. Below that, the material is not ready to be built on, and any forward progress that day is progress that will have to be made a second time — which is how a class arrives at Theorem 16 holding thirty percent of Theorem 3.

The expanding schedule. Anything newly firmed comes back at widening intervals. Once an item has cleared all five returns, it is his.

Return	When	At what tier
1st	Next session	Tier 3 — reconstruct from the diagram
2nd	Three days later	Tier 2 — the way, and how it is proved, orally
3rd	One week later	Tier 3 again
4th	Three weeks later	Tier 1 or 2, in the warm-up
5th	Two months later	Tier 4 if it is a keystone; otherwise Tier 3

4.5 Two students, three years apart

The twelve-year-old will be faster at nearly everything for the next few years. If he is also always first, the nine-year-old will quietly learn that his job in this class is to wait, and that lesson is far more durable than any theorem. Four countermeasures, all cheap:

1. **Unison first, every time.** He answers every question out loud before anyone can beat him to it.
2. **Differentiate the output, never the content.** Same theorem, same reasons, same standard of citation. The twelve-year-old writes from a blank page; the nine-year-old writes on a skeleton with the reasons blanked. Reduce his support across the twelve days — reason-blanks, then step-blanks, then blank. Nobody gets easier geometry; one gets more support in the writing, which is a different thing and can be said out loud without shame.
3. **Give the older boy the skeptic's chair rather than more speed.** Being the objector in a disputation (Drill 8) is genuinely harder than being the prover, and it costs the younger boy nothing. Extension for the twelve-year-old should always be *depth or rigour*, never *more theorems*, or you will end up running two classes.
4. **Watch for the ceiling being the writing, not the geometry.** At nine, the constraint is very often handwriting stamina and spelling, not reasoning. If he can give you Theorem 8 orally, complete, with every citation, he knows Theorem 8. Let him dictate it to you once in a while and write it out himself the next day.

Part Five • The Drill Menu

Nine formats. Each is short, each targets one tier, and each can be run with nothing but the cards, the wall chart, and a whiteboard. Part Six tells you which to run on which day; this part is the reference for how each one goes.

Drill 1 • CITE-IT • Layer 1 • 2-3 minutes • every single session

The highest-value drill in the packet. You describe a configuration; they name the authority. Speed is the point — twenty items in two minutes. Use the Givens cards.

T: "Two lines from the centre of a circle to the edge. What do we know?" [pause • signal]
 SS: "They're equal!"
 T: "How do we know?" [signal]
 SS: "Definition seventeen!"
 T: "Yes. Definition seventeen. • Two things each equal to a third thing." [signal]
 SS: "Common Notion one!"
 T: "A whole and one of its parts." [signal]
 SS: "Common Notion five — the whole is greater!"
 T: "Two triangles I can lay exactly on top of each other." [signal]
 SS: "Common Notion four!"

Run it standing up if you like. It is the drill that fits in seven minutes in a kitchen, and it is the one that most directly repairs losing the place.

Drill 2 • TWO-WAY THEOREM DECK • Tier 1 • 3 minutes

Both directions, always. Number — statement, and statement — number. The second direction is the one that matters and the one most decks never train, because it is what you actually do inside a proof: you recognise that you need *this fact* and must retrieve its label.

T: "Theorem six." [signal] SS: "Side-Side-Side!"
 T: "Full statement." [signal] SS: "If the three sides of one triangle equal the three sides of another, the corresponding angles are equal and the areas are equal."
 T: "Which theorem lets me say two triangles are equal when I know a side, the angle, and the next side?" [signal]
 SS: "Theorem two — SAS!"

Drill 3 • WHICH WAY OF DEMONSTRATING? • Tier 2 • 2 minutes • discrimination

Read a statement — nothing else, no diagram — and they name the way of demonstrating, and how it is proved. This is *discrimination* practice: the difficulty in real proof-making is almost never carrying out an operation, it is knowing which kind of argument the statement calls for. Mix the ways deliberately; never run five of the same in a row, or they will follow the sequence instead of the statement.

T: “In any triangle, a greater angle has opposite it a greater side. Which way?” [signal]
 SS: “Trichotomy elimination!” T: “Proved by what?” [signal] SS: “Theorem 3 and Theorem 15!”
 T: “Given a straight line, construct its midpoint. Which way?” [signal]
 SS: “Construct and verify!” T: “Verified by?” [signal] SS: “SAS!”

Drill 4 • WHERE ARE WE? • the wall chart • 1 minute • every session

At the wall chart. Three questions, always the same three:

- “Point to the theorem we are proving today.”
- “Point to everything it stands on.”
- “Point to something that will stand on it.”

One minute a day. It is the direct antidote to “they lose the place,” because a boy who can answer the third question knows what the proof is *for*.

Drill 5 • FADING THE SUPPORT • Tier 3–4 • written • 10 minutes

The written demonstration, at one of three levels of support. Take each boy down one level at a time, independently of the other, and only when the level above is at 100%.

Level	What is on the page	Drop to the next level when
A • Reason blanks	All steps printed in order; the citation column blank	He fills every citation correctly twice running
B • Step blanks	Citations printed; the steps blank	He reconstructs every step twice running
C • Blank slate	The statement and a blank page. No diagram supplied — he draws it.	This is Tier 4. It is the destination.

Drill 6 • SCRAMBLE • Tier 3 • 4 minutes • especially good for the nine-year-old

Cut a proof’s steps into strips; he reassembles them in order and then reads them aloud with citations. It isolates the *architecture* of the argument from the burden of producing the sentences, which for a nine-year-old is often the real bottleneck. Make it harder by including one strip that does not belong.

Drill 7 • DIAGRAM-ONLY RECONSTRUCTION • Tier 3 • 5 minutes • the core drill

Show the finished diagram and nothing else. He gives the whole proof orally, citing every reason. This is the drill that most directly produces what you are after, and it should appear in almost every session in the permanent warm-up.

Your only permitted interventions: “**Reason?**” when a step is asserted bare, and “**What are you trying to show?**” when he has genuinely lost the thread. Everything else is silence. If he cannot recover after

those two prompts, stop the test and re-model — the item was not firm and testing it further only rehearses the failure.

Drill 8 • THE SKEPTIC • once a week • 10 minutes • the disputation

One boy demonstrates; you — or, better, the twelve-year-old — play the objector. Run it in the scholastic form they are already learning: *videtur quod non*, an objection; *sed contra*, the authority against; *respondeo*, the demonstration; then the reply to the objection.

Good standing objections, all of which have real answers in the text:

- “You said the two circles meet at C. Which postulate says they meet?” (See the box below — this one is not answerable, and that is the point.)
- “You proved AB is not greater than AC. Perhaps AC is greater than AB.” (Theorem 4 — the second case.)
- “You used Theorem 15 to prove Theorem 16. Did you not use 16 to prove 15?” (Make him show the chart. This is where the map earns its keep.)
- “Why may you lay one triangle on top of another at all? Who says figures can be moved?” (Common Notion 4 — and it is a real philosophical question, which Euclid's superposition proof has always attracted.)

This drill is where the geometry and the logic course become one course. It is also, in my experience of what twelve-year-olds enjoy, the thing they will ask to do again.

Drill 9 • ERROR HUNT • rigour • 5 minutes

Hand him a written proof with a planted flaw. He must find it and name what kind of flaw it is. Three kinds worth planting, in rising order of difficulty:

5. **Out of order.** A proof of Theorem 14 that cites Theorem 28 (the angle sum) — perfectly true, and unavailable. Ask: *is it false, or is it early?* The distinction is worth a whole conversation.
6. **Illicit conversion.** Inside a proof of Theorem 3, a step that concludes a triangle is isosceles from its equal base angles. True — but it is Theorem 4, which has not been proved yet, and which was proved *using* Theorem 3. Circular.
7. **A bare assertion.** A step with no citation at all, buried among steps that have them. Surprisingly hard to spot, and exactly the habit you are trying to prevent.

A REAL GAP TO HAND THE TWELVE-YEAR-OLD

In Theorem 1, step 3 says: *these two circles intersect each other at points C and D.*

Nothing in the definitions, postulates or common notions guarantees that they do. Postulate 3 lets you draw each circle; nothing says two drawn circles must meet. Euclid assumes it, every reader for two thousand years let him, and it took the nineteenth century to notice that a continuity principle has to be added to make the very first proposition of the *Elements* airtight.

This is not a planted error — it is a genuine and famous gap, and handing it to a bright twelve-year-old as a real discovery rather than a puzzle you set him is worth more than a month of exercises. It teaches, at exactly the right age, that the authorities are a *finite list*, that a step is legitimate only if something on the list licenses it, and that this remains true even for Euclid.

Part Six • The Twelve-Day Consolidation Block

Twelve sessions. At four or five a week that is two and a half to three weeks. Theorem 17 waits — and the two and a half weeks will be repaid several times over, because everything from Theorem 17 to Theorem 37 leans on this material, and Chapter 1 ends at Pythagoras.

THE STANDARD SHAPE OF A FULL SESSION (45 MINUTES)

0–5 • Warm-up. Drill 1 (Cite-It) + Drill 4 (Where are we?). Never skipped, in any session, ever.

5–12 • Component firming. Whatever Layer 1 or Layer 2 items today's proofs will need. You are pre-loading, so that nothing has to be retrieved under pressure during the demonstration.

12–30 • The focus proofs. Model — Lead — Test, in that order, without blending them.

30–40 • Written demonstration. Drill 5 (fading the support), each boy at his own level.

40–45 • Delayed check + log. Yesterday's item, cold. Then one line in the log: what firmed, what failed, tomorrow's first item.

Day 1 • The Givens and the order of dependence — no proofs at all

Objective. Get the fourteen working givens to reflex speed and put the whole chapter on the wall. A day with no theorem in it feels like a day lost; it is the highest-yield day in the block.

Segment	Minutes	What you run
Warm-up	5	Introduce the signal. Explain that answers now come in unison. Practise the signal on three easy items.
Firming	20	Drill 1 on the fourteen working givens (Part 3.1), in both directions: item $\rightarrow$ what it licenses, and configuration $\rightarrow$ item. Aim for sixty responses.
The chart	15	Put up the wall chart. Name the four stages aloud. Then Drill 4 on eight different theorems: “point to what it stands on.”
Close	5	Two cards each to take away: Def. 17 and C.N. 5. “These two sentences build most of Chapter One.”

If it goes badly: it will not. This day is easy on purpose — it re-establishes success after a stretch in which they have felt half-lost. Note in the log which givens were slow; those become the first items of every warm-up for the rest of the block.

Day 2 • Theorems 1–2 • Equality without measuring

Objective. Establish the two ideas the whole chapter runs on: a circle gives equality for free, and superposition gives SAS.

Segment	Minutes	What you run
Warm-up	5	Drill 1 (givens) + Drill 4
Firming	7	Post. 3, Def. 17, C.N. 1, C.N. 4, and the principle that two straight lines cannot enclose a space. Twenty responses.
Focus	18	Model-Lead-Test on Theorem 1, then Theorem 2. For Theorem 2 the load-bearing step is why AC <i>must</i> fall on DF — make them give the two-lines principle unprompted.
Written	10	Both boys: Theorem 1. Twelve-year-old at Level B (step blanks); nine-year-old at Level A (reason blanks).
Close	5	Delayed check on three of yesterday's slow givens. Log.

If it goes badly: the usual failure is Theorem 2 — superposition feels like cheating, and a bright child will object to it. Do not suppress the objection; it is a good one and philosophers have made it for two thousand years. Answer with C.N. 4, then move on. Understanding the *discomfort* of superposition is worth more than a smooth day.

Day 3 • Theorems 3-4 • The isosceles pair, and the first reductio

Objective. Get the two-congruences-then-subtract structure of Theorem 3, and introduce Way C explicitly by name.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	C.N. 2, C.N. 3, C.N. 5, Def. 20. Then Drill 3 (which way of demonstrating?) on four statements — this is where you first teach the five names.
Focus	20	Theorem 3 first: model it twice, because it holds two congruences and a subtraction. Then Theorem 4: name Way C aloud before you start, and say what the shape of the argument will be before you prove it.
Written	8	Theorem 3, Level A for both — the citation load is heavy here.
Close	5	Delayed: Theorem 1, cold, from the diagram (Drill 7). Log.

Say this out loud before Theorem 4: “Theorem 3 says isosceles gives equal base angles. Now we want the reverse. Does the reverse come free?” Let them answer. It is the same question as simple conversion of an A-proposition in the logic course, and this is the moment to say so.

If it goes badly: Theorem 3 is genuinely hard — the first proof with overlapping triangles. If the Test phase collapses, drop the written work and re-model. Never let Day 3 run long into Day 4's material.

Day 4 • Theorems 5–6 • The three sides determine the angles, and SSS

Objective. See that SSS is not proved directly: it follows from Theorem 5.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	Drill 2 (two-way deck) on Theorems 1–4. Both directions.
Focus	18	Theorem 5, then Theorem 6. Ask before Theorem 6: “We want SSS. What do we already have that would give it to us?” Let them find Theorem 5 on the chart.
Written	10	Theorem 4 (reductio) — Level A for the nine-year-old, Level B for the twelve-year-old.
Close	5	Delayed: Theorem 3, cold. Log.

If it goes badly: the substitution in Theorem 5 (replacing $\angle 2$ by $\angle 3 + \angle 4$) is where they lose it. Write the two equations on separate lines and physically point at the substitution. It is an algebraic step inside a geometric proof, and it is the first one they have met.

Day 5 • Checkpoint A • Theorems 1–6

Objective. Measure, then reteach whatever failed. This is a measurement day, not a teaching day — resist the urge to help.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Checkpoint	20	Checkpoint A, in full (Part 8.1). Score it honestly as you go.
Reteach	15	Whatever failed, Model-Lead-Test again. Nothing else.
Close	5	Log, with the failed items named. They become the first items of every warm-up until they clear.

The rule for this day: if two or more sections of the Checkpoint fail, do not proceed to Day 6. Repeat Day 5 with the failed material as a whole extra session. A block that takes fourteen days and produces mastery is a success; a block that takes twelve and produces sixty percent has reproduced the problem you started with.

Day 6 • Theorems 7–8 • Bisection

Objective. See both bisections as one idea — put the equilateral triangle on it, then discover a congruence.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	Operation 5 (erect the equilateral triangle) and Operation 7 (discover a congruence) by name. Drill 3 on four Way A statements.
Focus	18	Theorems 7 and 8 together. Make the parallel explicit: both start with Theorem 1. Theorem 7 verifies with SSS ; Theorem 8 verifies with SAS . That difference is the single most commonly miscited fact in the chapter.
Written	10	Theorem 7, Level B / Level A.
Close	5	Delayed: Theorem 6, cold. Log.

Watch for: in Theorem 7 they forget that BF is the third pair of equal sides (it is common to both triangles). A side that is shared rather than constructed is invisible to a beginner. Say “common side” every single time, in every proof that has one — it recurs in 8, 9, 10 and 14.

Day 7 • Theorems 9-10 • The two perpendiculars

Objective. See 9 and 10 as one construction under two conditions — the point on the line, and the point off it.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	Def. 9 and Def. 10 to reflex. These two are where students stop one step short.
Focus	18	Theorems 9 and 10. Ask first: “What is different about the two cases?” Then model both. Note that 9 uses Thm. 1 and 10 uses Thm. 8 — different scaffolds, same finish.
Written	10	Theorem 9, Level B / Level A.
Close	5	Delayed: Theorem 7, cold. Log.

The step they will skip, in both: having proved the two angles equal by SSS, they stop. But “equal” is not “right.” Definition 9 is what converts two equal adjacent angles into two right angles, and Definition 10 is what converts that into perpendicularity. Two extra citations, both omitted by almost everyone. Drill them as a pair: “Equal adjacent angles — so?” “Right angles, Definition 9!” “And therefore?” “Perpendicular, Definition 10!”

Day 8 • Theorems 11-13 • Two right angles as a unit of account

Objective. Way D: finding what is equal. Light on construction, heavy on C.N. 1 and C.N. 3.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	C.N. 1 and C.N. 3, fast. Then Post. 4 — which appears exactly once in the whole chapter, in Theorem 12, and is worth flagging for that reason.
Focus	20	All three. Theorem 11 (note: the perpendicular is <i>constructed</i> by Thm. 9, not given), Theorem 12 (Way C again — say so), Theorem 13 (easy; use it to end on success).
Written	8	Theorem 13, Level C for the twelve-year-old — his first blank-slate proof. Level B for the nine-year-old.
Close	5	Delayed: Theorem 9, cold. Log.

Use Theorem 13 deliberately as a confidence day. It is four lines long and follows from Theorem 11 twice. After a hard week, a proof a boy can produce whole and unassisted is worth building the day around.

Day 9 • Checkpoint B • Theorems 7-13

Objective. Measure, then reteach. Same rules as Day 5.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Checkpoint	20	Checkpoint B, in full (Part 8.2).
Reteach	15	Whatever failed.
Close	5	Log.

Same rule as Day 5: two or more failed sections means Day 9 repeats. Theorem 14 is the hardest proof so far and it stands on Theorems 8 and 13. Do not attempt it on soft ground.

Day 10 • Theorem 14 alone • the keystone

Objective. One theorem, one whole session. This is the proof your block exists to secure.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4 — and today put a finger on 14 and trace both arrows down from it to 15 and 16.
Firming	8	Rehearse the four components separately before you assemble anything: bisect a line (Thm. 8), vertical angles (Thm. 13), SAS (Thm. 2), whole > part (C.N. 5). Each on its own, to fluency.
Focus	22	Model it three times. Once slowly, naming the four components as you use them. Once at normal speed. Once with the boys leading in unison. Only then Test.
Written	5	Nothing new. Level A on Theorem 14 — the reasons only.
Close	5	Delayed: Theorem 11, cold. Log.

Why this is hard, said plainly to them: every proof before it used one or two tools. This one uses four, and then repeats itself for the second remote angle. Tell them that in advance. A student who knows a proof is long is patient with it; a student who expected four lines and is on line nine concludes he is lost.

If it goes badly: it is entirely acceptable to give Theorem 14 two sessions. It is the only theorem in 1–16 of which that is true.

Day 11 • Theorems 15–16 • Inequality, and the elimination proof

Objective. Way E, and the third converse pair — what everything before it was for.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Firming	7	Drill 3 on all five ways, mixed. Then Operation 4 (cut off an equal length) — Theorem 15 opens with it.
Focus	18	Theorem 15, then Theorem 16. Before 16: “This is the converse of 15. What did the last two converses cost us?” Let them predict indirect proof. Then show them trichotomy — three cases, kill two.
Written	10	Theorem 16, Level C for the twelve-year-old, Level B for the nine-year-old. It is short, and it is the most <i>logical</i> proof in the chapter — a good one to own outright.
Close	5	Delayed: Theorem 14, cold. Log.

The insight to name aloud: Theorem 16 contains no construction whatsoever. Nothing is drawn, nothing is cut off, nothing is joined. It is pure reasoning from Theorems 3 and 15. After fifteen proofs of drawing circles, that is a genuine surprise, and it is the first glimpse of what the rest of mathematics is going to feel like.

Day 12 • Checkpoint C, the cumulative, and a disputation

Objective. Measure everything; then finish the block with something they will want to do again.

Segment	Minutes	What you run
Warm-up	5	Drill 1 + Drill 4
Checkpoint	12	Checkpoint C (Part 8.3) — Theorems 14–16.
Cumulative	13	The cumulative check (Part 8.4): six theorems drawn at random from all sixteen, Tier 3, oral, from the diagram.
Disputation	13	Drill 8. Twelve-year-old objects; nine-year-old demonstrates; then swap. Full scholastic form.
Close	2	Log. Then say plainly what they have done: sixteen demonstrations, owned.

Then stop and mark it. The end of a block should be visible — a certificate, a signature at the bottom of the wall chart, whatever suits your house. The point is that they experience *consolidation* as an accomplishment in its own right, and not as the price of having forgotten things. That framing will matter every time you run a block for the rest of the course.

Part Seven · The Permanent Ten-Minute Warm-up

The block fixes what has already decayed. This fixes the decay itself, and it is the part that must survive the block. From Day 13 onward — and for the whole of Chapters 2, 3 and 4 — every session begins with these ten minutes before any new material is touched. If a session is too short for anything else, this is the session.

7.1 The four slots

Slot	Min	What	Which drill
1 · The givens	2	The fourteen working givens, plus any that were slow yesterday	Drill 1 · Cite-It
2 · Yesterday	2	Yesterday's theorem — the way, and how it is proved, orally	Drill 3 · Which way?
3 · The rotation	5	One older theorem off the schedule below, reconstructed from its bare diagram	Drill 7 · Diagram-only
4 · The chart	1	Where are we, what do we stand on, what will stand on us	Drill 4 · Where are we?

Slot 3 is what produces long-term retention, and it is the one to protect when time is short. Two minutes of genuine reconstruction beats ten minutes of re-reading, every time — and re-reading is what will happen by default if you do not have a named drill in the slot.

7.2 The rotation

Slot 3 works through the sixteen on a simple cycle, weighted so the keystones come round more often. Add each new theorem to the cycle as you prove it; the cycle simply gets longer, and each theorem comes round less often as the chapter grows — which is exactly the expanding schedule you want, achieved for free.

Week	Slot-3 theorems, one per session
A	1 · 3 · 2 · 8 · 5
B	6 · 4 · 11 · 14 · 7
C	2 · 9 · 13 · 16 · 1
D	14 · 10 · 8 · 12 · 15
E	11 · 6 · 3 · 14 · 2
then	repeat, inserting new theorems as they are proved

Keystones 1, 2, 6, 8, 11 and 14 appear two or three times in the five-week cycle; the rest appear once. Do not over-engineer this — the schedule only has to be *written down and followed*, and any reasonable weighting will beat “whatever I thought of this morning.”

7.3 Re-entry after a break

This is the single highest-leverage rule in the packet for a homeschool calendar, and it costs nothing but the discipline not to resume where you stopped.

THE RE-ENTRY PROTOCOL

Never resume forward progress on the first day back. Not after a holiday, not after illness, not after a fortnight in which the geometry simply did not happen.

One week away — one session of pure review before new material.

Two to four weeks — two sessions of review, and re-run Slot 3 on the three theorems most recently proved.

Longer than a month — three sessions, ending with the relevant Checkpoint from Part Eight. Treat the Checkpoint result as real; if it fails, run a short block rather than pressing on.

The instinct after a break is always to make up lost ground. It is exactly wrong: what was lost was not ground, it was firmness, and pressing forward on unfirm material is how a curriculum arrives at Chapter 4 with a class that cannot prove Theorem 8.

Part Eight • The Mastery Checks

Four checks. Each is short enough to run inside a session, and each ends in a decision — advance, or repeat. The decision is the point; a check that never blocks progress is not a check.

8.1 Checkpoint A • Theorems 1–6 • Day 5 • 20 minutes

#	The task	Pass mark
1	Cite-It, 20 items drawn from the fourteen working givens, oral	18 of 20, each inside a second
2	Two-way deck, Theorems 1–6, both directions	6 of 6, both ways
3	Which way of demonstrating? — 6 statements, no diagrams	5 of 6, with the proof named too
4	Diagram-only reconstruction of Theorem 3 and Theorem 6, orally, every reason cited	Complete, with no more than two “Reason?” prompts each
5	Cold write: Theorem 1 (both boys) and Theorem 4 — Level C for the twelve-year-old, Level A for the nine-year-old	Every step present, every citation correct

8.2 Checkpoint B • Theorems 7–13 • Day 9 • 20 minutes

#	The task	Pass mark
1	Cite-It, 20 items, weighted toward Def. 9, Def. 10, C.N. 1, C.N. 3, Post. 4	18 of 20
2	Two-way deck, Theorems 1–13	12 of 13, both ways
3	“Theorem 7 verifies with which test? Theorem 8?”	SSS and SAS, unhesitating
4	Diagram-only reconstruction of Theorem 8 and Theorem 11	Complete, ≤ 2 prompts each
5	Cold write: Theorem 13 (Level C, both) and Theorem 9 (Level B / Level A)	Every step, every citation — including Def. 9 and Def. 10 at the end of Theorem 9

8.3 Checkpoint C • Theorems 14–16 • Day 12 • 12 minutes

#	The task	Pass mark
1	“Theorem 14 uses four earlier results. Name them.”	Thms. 8, 13, 2 and C.N. 5 — all four
2	Diagram-only reconstruction of Theorem 14	Complete, ≤ 3 prompts
3	Which way of demonstrating, for 15 and for 16, with the difference explained	Both, plus “16 has no construction”

#	The task	Pass mark
4	Cold write: Theorem 16	All three cases named, two eliminated with the right authorities

8.4 The cumulative check • Day 12 • 13 minutes

Draw six of the sixteen at random — genuinely at random; write the numbers on slips. For each: the bare diagram, oral reconstruction, every reason cited. Alternate boys.

Pass: five of six complete with no more than two prompts each. **This is the number that tells you whether the block worked.** Everything else in Part Eight measures a segment; this measures the thing you set out to build.

8.5 What to do with the result

Result	What it means	What you do
All sections pass	Chapter 1's foundation is firm through Theorem 16	Resume at Theorem 17 next session, with the Part Seven warm-up permanently in place
One section fails	A local gap	One extra session on that material, then re-check only that section. Then resume.
Two or more fail	The block was too fast, not the boys too slow	Repeat the relevant days. Do not resume forward progress. This is the correct outcome to accept gracefully — it is what the check is for.
Everything passes easily and fast	You underestimated them	Resume, and lengthen the Slot-3 rotation so old material comes round less often. Do not skip the warm-up on the strength of one good result.

Part Nine · Running This as a Father, Not as a Department

The rest of this packet is about geometry. This part is about the conditions under which any of it survives a real year with a real family, and it is the part I would read twice.

9.1 Coverage and mastery — choose on purpose

A school curriculum optimises coverage. It has to: twenty-five students, a fixed calendar, an examination in the spring, and a syllabus written by someone who will never meet the class. Coverage is the rational objective under those constraints, and retention is what gets traded away to buy it.

You have two students, no calendar, and no examination. Your comparative advantage is precisely that you are free to decline coverage in favour of mastery — and if you do not decline it deliberately, you will drift into a school's pace and inherit a school's retention, which is the thing you are already unhappy about.

WRITE THIS DOWN SOMEWHERE THE BOYS CAN SEE IT

We will finish Chapter 1 when Chapter 1 is firm.

Then stop tracking pace. Not loosely — actually stop. Do not keep a private count of where you “should” be, because that count has no referent: it was computed for a class that does not exist, on a calendar you do not keep, toward a test nobody will sit.

9.2 The two non-negotiables

Everything else in this packet can flex. These two cannot, and holding them is what makes flexing everything else safe.

8. **The sessions happen often, even when they are short.** Frequency, not duration, builds retention. Five short sessions in a week beat one long one, and it is not close.
9. **Nothing new is built on something unfirm.** When the 90/100 criterion fails, the day's job is the old material, whatever the plan said.

The second rule is what makes the first one survivable. If you never build on what is not yet firm, then a short day is never a wasted day — it is simply a day that firmed rather than advanced, and firming is real work with a real product.

9.3 Three session tiers — decide them now, not in the moment

The reason lessons get cancelled is that the only available option is the full one. Pre-define three, so that a bad day degrades gracefully instead of disappearing.

Tier	Time	What it contains	When
FULL	45 min	The whole structure: warm-up, firming, Model-Lead-Test, written demonstration, delayed check, log	A normal day

Tier	Time	What it contains	When
SHORT	20 min	Warm-up (10) + one Model-Lead-Test on one proof (10). No writing at all.	A busy day, or a day you are tired
MINIMUM	7 min	The warm-up alone — Cite-It, yesterday's theorem, one map placement. Standing in the kitchen counts.	A day that was going to be cancelled

The Minimum session is not a token and it is not a consolation. It keeps the sessions unbroken and the givens at reflex speed, which is well over half the value of the whole system. Two Full sessions and three Minimums in a week will outperform one heroic ninety-minute Saturday — and will also leave two boys who think of geometry as something that is simply always going on, rather than an event that requires a clear afternoon.

A practical note: name the tiers out loud, to them. “Today is a Minimum.” A named short session is a session. An unnamed one is a failure to have class.

9.4 Father and professor — keeping the roles clean

You are doing something a professor is not asked to do: correcting, hour after hour, a person whose good opinion of himself is partly your responsibility. That is a real difficulty and it is better managed structurally than by willpower.

- **Mark the boundary.** Same place, same opening words, a definite end. “Class is in session.” ... “That is the end of class.” The point is not ceremony. It is that correction inside a marked boundary is not a father correcting a son; it is a teacher correcting an answer — and children can tell the difference, if you make the boundary visible enough for them to see it.
- **Let the script do the correcting.** This is the underrated benefit of Direct Instruction in a home classroom. When correction is a fixed five-step procedure, it carries no displeasure with it. You are not disappointed; you are carrying out step two. Neither of you has to interpret anything.
- **Never carry the class into the evening.** If a session went badly, it ended when it ended. Do not reopen it at dinner. A boy who suspects that a bad class will follow him to the table will stop offering wrong answers — and wrong answers, offered freely, are the entire diagnostic instrument you have.
- **Do not teach depleted.** Run a Minimum instead. Seven minutes from a calm father is worth more than forty-five from a frayed one, and the cost of the frayed session is not the lost time — it is the association, which is expensive and slow to repair.
- **Expect one specific distortion.** You will read ordinary confusion as a verdict on your parenting. It is not. Confusion is data, it is the normal output of a class that is aimed correctly, and a session in which nobody was confused was probably a session in which nobody learned anything.

9.5 What “behind” would even mean

There is no track. Their coevals are not doing Euclid; the comparison class is empty. If you want a benchmark, make it a retention benchmark rather than a pace one:

“Can he demonstrate Theorem 8, cold, six months from now?”

That is a real standard. It is harder than any school's, it is the one this method is actually built to hit, and it has the further advantage of being answerable — you can simply ask him in February.

It is also worth saying plainly, on the days when it does not feel like it is going well: two boys of nine and twelve who are writing out Euclidean demonstrations by hand are doing something most undergraduates never do. The thirty percent you are worried about is thirty percent of material that most people never encounter at all. The gap you are trying to close is the gap between very good and excellent, which is a much better problem than the one you might have had.

9.6 The one-line log

Thirty seconds at the end of every session, including Minimums. Five fields.

Date	Tier	Firmed	Failed	Tomorrow's first item
F / S / M				

It does three jobs. It makes step 5 of the firming cycle actually happen across days — that step needs a written record or it does not exist. It gives you honest evidence on the days you feel behind, which is usually the evidence you most need and least have. And if your state ever asks for a portfolio, you already have one, produced at no extra cost.

9.7 Where the app belongs

Your instinct to keep it out of the classroom is right, and the sharper reason is worth having explicitly: **the app supplies the steps, and supplying the steps removes exactly the retrieval that produces retention.** Recognising a proof as correct and being able to generate it are different capacities, and only one of them is what you are after. An hour in a well-built app can feel like mastery and produce almost none.

But it has three excellent jobs outside the demonstration itself, and it would be a waste not to use it for them:

- **As your answer key and prep tool.** Building the day's Cite-It items, checking a citation before you correct a boy on it, confirming which theorem a proof leans on.
- **As the delayed self-check.** After a boy has written a proof cold, on paper, he opens the app and marks himself against it. Grading your own work against an authority is a completely different act from being shown the answer, and it is a good one.
- **For the animated constructions, after the argument is owned.** Watching the circles swing once the proof is already his is a pleasure and costs nothing. Watching them before is a shortcut that costs the proof.

And use the theorem sheets you have already built as the flash decks — you have the exporter. **Never during a first attempt.**

9.8 What to do this week

10. Print the wall chart and put it up where the boys can reach it.
11. Print and cut the card decks. Start with the fourteen working givens; the rest can wait.
12. Tell them what is happening and why: “We are going back over the first sixteen so that we own them, and then we go on.” Frame it as consolidation, not as remediation — it is the truth, and the difference in how they receive it is large.
13. Run Day 1. It is the easy one, and it will go well.
14. Write the first log line. That is the habit that carries all of the rest.

Appendix • Theorems 1–16 at a Glance

For your own use while building drills, and for a fast check before a session. **Way of demonstrating:** A = construct and verify · B = discover congruences · C = reductio · D = find what is equal · E = trichotomy elimination.

#	Theorem	Way	How it is proved	The step they will skip
1	Equilateral triangle on a given line	A	Two circles (Post. 3 + Def. 17), chained by C.N. 1	They draw it and stop. The proof is the three equalities, not the picture.
2	SAS	B	Superposition (C.N. 4) + two lines cannot enclose a space	Why AC <i>must</i> fall on DF. Without the two-lines principle the proof has a hole.
3	Isosceles $\Rightarrow$ equal base angles	B	SAS twice, then C.N. 3	There are two congruences and then a subtraction. They do one and stop.
4	Equal base angles $\Rightarrow$ isosceles	C	Reductio landing on C.N. 5	The second case. Ruling out $AB > AC$ is only half of it.
5	The three sides determine the angles — triangles are rigid	C	Thm. 3 twice, then substitute, then C.N. 5	The substitution — replacing $\angle 2$ by $(\angle 3 + \angle 4)$. It is algebra inside geometry.
6	SSS	B	Follows from Theorem 5, then C.N. 4	They try to prove it directly. It is not proved directly; it follows from Theorem 5.
7	Bisect an angle	A	Thm. 1, verified by SSS	BF is the third pair of sides — the common side. Shared sides are invisible to beginners.
8	Bisect a line	A	Thm. 1 + Thm. 7, verified by SAS	They cite SSS. Theorem 7 uses SSS; Theorem 8 uses SAS. The most-miscited fact in the chapter.
9	Perpendicular from a point <i>on</i> a line	A	Thm. 1 + SSS, then Def. 9 and Def. 10	Stopping at “the angles are equal.” Equal is not right. Def. 9, then Def. 10.
10	Perpendicular from a point <i>off</i> a line	A	Thm. 8 + SSS, then Def. 9 and Def. 10	Forgetting to bisect GE — the bisection is what makes the two triangles congruent.
11	Adjacent angles = two right angles	D	Thm. 9 (constructed!), Def. 9, C.N. 1	Treating the perpendicular as given. It is constructed, by Theorem 9.
12	Two right angles $\Rightarrow$ straight line	C	Thm. 11 + Post. 4, landing on C.N. 5	Post. 4 — the one place in the chapter where it is essential, and always omitted.
13	Vertical angles are equal	D	Thm. 11 twice, then C.N. 3	Nothing. This is the clean one — use it to end a hard week on a success.

#	Theorem	Way	How it is proved	The step they will skip
14	Exterior angle > remote interior	B	Thm. 8 + Thm. 13 + SAS, finishing on C.N. 5	All of it, and then the <i>second half</i> for the other remote angle. Model three times.
15	Greater side $\Rightarrow$ greater angle	B/C	Cut off (Operation 4) + Thm. 3 + Thm. 14 + C.N. 5	Cutting off on the wrong segment. AD = AB is laid off along AC, the greater one.
16	Greater angle $\Rightarrow$ greater side	E	Thm. 3 and Thm. 15, by elimination	Attempting a construction. There is none — the whole proof is three cases and two kills.

READ THE LAST COLUMN ONCE BEFORE EVERY SESSION

It is a list of sixteen specific, predictable omissions. Almost every wrong or incomplete demonstration your boys produce in Chapter 1 will be on it. Knowing the error in advance turns correction from a diagnosis into a reflex — which is the whole idea, applied to you rather than to them.